Agent-Based Simulation of Trust Development in Human-Robot Teams

Abstract

Human-robot teams are increasingly deployed in complex operational environments, yet validated tools to predict team performance under varying conditions remain scarce. This paper presents an agent-based model, implemented in NetLogo, that captures trust dynamics, workload distribution, and performance in human-robot teams of 3–6 agents performing collaborative tasks. The model implements a trust-dependent delegation mechanism in which delegation probability for judgment tasks follows a quadratic function of trust, creating an over-reliance pathway at high trust levels. An adaptive Kalman trust filter with stress- and transparency-modulated measurement noise governs trust evolution, producing calibrated trust trajectories ($R^2 = 0.850$, compared with Hancock et al., 2021 benchmarks). A dedicated reliability sweep (15 levels, 750 runs) reveals a non-linear reliability-performance relationship best characterized by a cubic polynomial ($\Delta AIC = 79$ over linear; adjusted $R^2 = 0.848$), with performance peaking at 95% reliability before declining at 100%. The mechanistic pathway involves trust-mediated delegation of judgment-critical tasks to robots, leading to a systematic misallocation that increases from 4.7% to 28.8% across the reliability range. A crossed reliability × transparency experiment (1,250 runs) reveals a metric-dependent reversal of predictor dominance: reliability drives task success rate ($\eta^2 = 0.842$) while transparency drives misallocation rate ($\eta^2 = 0.090$), yet transparency has no detectable effect on aggregate performance ($\eta^2 < 0.001$). This finding demonstrates that standard performance metrics may mask differences in allocation quality, with consequences for safety and human skill maintenance.

Keywords: human-robot interaction, trust dynamics, agent-based modeling, over-reliance, trust-dependent delegation, misallocation, Kalman filter

1. Introduction

The integration of robots into human work teams represents a fundamental shift from automated tools to collaborative partners. Unlike traditional automation, in which humans supervise machines, modern human-robot collaboration requires continuous interaction, mutual adaptation, and the development of trust (Hancock et al., 2011). This shift is evident in manufacturing, healthcare, and defense operations, where robots must adapt to dynamic human behaviors and uncertain task requirements.

Despite growing deployment, predictive tools for designing effective human-robot teams remain limited. Practitioners rely on trial-and-error approaches, risking team failure and the erosion of trust. While empirical studies have established that robot reliability is the strongest predictor of trust (Hancock et al., 2011, 2021) and that over-reliance on automation poses persistent risks (Parasuraman & Riley, 1997), computational models capable of reproducing team-level consequences of trust-mediated delegation decisions remain scarce. In particular, the conditions under which high trust produces appropriate versus inappropriate reliance — and whether standard evaluation metrics are sensitive enough to distinguish between the two — have received limited computational investigation.

This paper develops an agent-based model with trust-dependent delegation, calibrated against published experimental data, to explore these dynamics. The contributions include: (a) a simulation framework that produces emergent over-reliance dynamics through task-type-sensitive delegation functions, (b) identification of a non-linear reliability-performance relationship driven by judgment task misallocation, (c) demonstration that standard performance metrics mask transparency's substantial effect on allocation quality, and (d) an open-source implementation for practitioner use and community extension.

The distinction between calibration and validation is important to clarify at the outset. Calibration involves tuning model parameters so that outputs match known empirical results; validation involves testing model predictions against independent data not used during calibration (Sargent, 2013). This paper primarily reports calibration results supplemented by extensive sensitivity analysis. True out-of-sample validation against independent experimental data remains a priority for future work.

2. Background and Related Work

2.1 Trust in Human-Robot Interaction

Trust fundamentally determines whether humans will rely on robotic teammates. Lee and See (2004) define trust in automation as "the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability". This definition highlights goal alignment and uncertainty management as the two axes along which trust operates in human-robot teams.

2.1.1 Meta-Analytic Foundations

Three landmark meta-analyses provide the empirical foundation for computational trust modeling. Hancock et al.'s (2011) meta-analysis of 29 studies (69 correlational and 47 experimental effect sizes) identified robot performance as the strongest predictor of trust ($r = 0.71$), with environmental factors moderately associated and human-related factors showing limited effects. Their three-factor model — human-related, robot-related, and environmental factors — established the organizing framework adopted by subsequent research in the field. Hancock et al. (2021) updated this analysis with a substantially broader literature base, confirming that robot attributes and performance remained the dominant predictors while identifying additional individual precursors. Notably, expectancy ($r = 0.54$) and satisfaction ($r = 0.26$) emerged as significant human-related factors, and communication method and personality were identified as especially influential robot-related factors. The updated model (their Figure 7) expanded the constellation of trust antecedents while reinforcing the primacy of robot performance factors.

Schaefer et al. (2016) extended the meta-analytic foundation from robots to automation broadly, analyzing 30 studies providing 164 pairwise effect sizes and 16 studies providing 63 correlational effect sizes. They found a medium overall effect ($\bar{g} = 0.48$, $\bar{r} = 0.34$) with automation-related factors ($\bar{g} = 0.53$) exceeding human-related factors ($\bar{g} = 0.49$). Their finding that automation capability and reliability were the strongest automation-related predictors directly informed the present model's emphasis on reliability as the primary parameter.

2.1.2 Trust Models and Frameworks

Hoff and Bashir (2015) synthesized 101 empirical papers (127 eligible studies) into a three-layered trust model that distinguishes dispositional trust (stable individual tendencies shaped by personality and culture), situational trust (context-dependent factors, including system complexity and workload), and learned trust (experience-driven calibration through interaction).

Their framework is particularly relevant for simulation design because it specifies which trust components should be modeled as fixed agent properties (dispositional) versus dynamic state variables (learned). The present model implements dispositional trust through initial trust settings and learned trust through the adaptive Kalman filter update mechanism.

Lee and Moray (1992) provided the foundational dynamic trust model showing that trust tracks system performance over time — a relationship the present model reproduces through its Kalman filter convergence mechanism. Their autoregressive trust model demonstrated that trust at time t depends on both prior trust and observed performance, which is precisely the structure implemented in the adaptive Kalman formulation.

2.1.3 Computational Trust Dynamics

Guo and Yang (2021) developed a personalized trust prediction model using Bayesian inference, modeling trust as a Beta-distributed random variable updated through observed robot successes and failures. Their model adhered to three key properties: trust at time i is influenced by trust at time i−1, trust is more heavily impacted by negative experiences, and trust stabilizes with repeated interactions. They identified three distinct trust dynamic types — the Bayesian decision maker, the oscillator, and the disbeliever — achieving RMSE of 0.072 on a surveillance task dataset. The present model's Kalman filter formulation shares the Bayesian updating structure but extends it by incorporating stress- and transparency-modulated measurement noise, enabling investigation of contextual moderators of trust dynamics.

Xu and Dudek (2015) developed an online probabilistic trust inference model using a dynamic Bayesian network framework, treating trust as a hidden variable estimated from automation performance and agent behaviors. Their approach demonstrated that real-time trust estimation is feasible, thereby supporting the present model's tick-by-tick trust-updating architecture.

2.1.4 Trust Asymmetry

The observation that failures erode trust more severely than successes build it is consistently documented across empirical studies. Salem et al. (2015) demonstrated that behavioral inconsistency reduces trust more severely than performance failures alone ($d = 1.24$) in a human-robot cooperation paradigm. Slovic (1993) formalized the negativity bias in risk perception, arguing that trust is fragile — difficult to build and easy to destroy. Baumeister et al. (2001) provided the broader psychological framework, reviewing evidence across multiple domains that negative events have greater impact than positive events of equivalent magnitude. Tower and

Brooks (2024) found that proactive failure acknowledgment preserved substantially more trust than reactive explanations, suggesting that transparency about failures may buffer trust erosion. The present model implements a 5:1 trust asymmetry ratio (failure penalty to success gain), moderated by transparency, consistent with this literature.

Desai et al. (2013) examined the impact of robot failures and feedback on real-time trust, demonstrating that trust recovery after failures depends on both the type of failure and the feedback provided. Their finding that transparent feedback accelerates trust recovery informed the present model's transparency-modulated trust penalty mechanism.

2.2 Over-Reliance and Automation Misuse

2.2.1 Foundational Framework

Parasuraman and Riley's (1997) taxonomy of automation use problems — use, misuse, disuse, and abuse — remains the foundational framework for understanding over-reliance. Misuse, defined as over-reliance on automation resulting in monitoring failures and decision biases, is the central phenomenon the present model seeks to reproduce computationally. The authors identified three key factors affecting automation monitoring: workload (high workload increases misuse), automation reliability and consistency (high reliability paradoxically increases complacency), and the saliency of automation state indicators (i.e., transparency).

2.2.2 Automation Complacency

Parasuraman, Sheridan, and Wickens (2000) extended the foundational framework with a model for types and levels of human interaction with automation, distinguishing four functions (information acquisition, information analysis, decision selection, action implementation) at multiple levels of automation. Their framework predicts that higher levels of automation produce greater complacency risk — a prediction consistent with the present model's finding that performance plateaus at high reliability due to excessive delegation.

Parasuraman and Manzey (2010) provided a comprehensive review of automation complacency and bias, distinguishing complacency (insufficient monitoring of automated systems) from automation bias (using automated cues as heuristic replacements for vigilant information processing). Both phenomena are relevant to the present model: the delegation mechanism captures complacency-like behavior (delegating tasks without adequate scrutiny), while the transparency-based awareness check models the degree to which operators resist automation bias by evaluating task-robot fit.

2.2.3 Trust-Mediated Over-Reliance

The over-reliance pathway in human-robot teams operates through trust-mediated delegation. When trust is high, humans delegate more tasks to robots, including tasks for which human judgment would be more appropriate. This trust-reliance coupling is well-established: Lee and See (2004) demonstrated that trust and reliance, while related, are not identical — operators may trust automation yet choose not to rely on it, or rely on it without trusting it. The present model captures the trust-to-delegation pathway explicitly through task-type-sensitive delegation probabilities, enabling investigation of conditions under which trust leads to appropriate versus inappropriate reliance.

Robinette et al. (2016) demonstrated overtrust in emergency evacuation scenarios, where participants followed a robot's guidance even when it contradicted common sense. Their finding that overtrust persists even after experiencing robot failures underscores the importance of modeling trust-dependent delegation in safety-critical contexts.

2.3 Communication and Transparency

2.3.1 Transparency Frameworks

Chen et al.'s (2018) Situation Awareness-based Agent Transparency (SAT) model identifies three levels of transparency: Level 1 (what the agent is doing), Level 2 (why it is doing it), and Level 3 (projected outcomes). Each level provides progressively more information but also increases cognitive load, creating a transparency-workload tradeoff. The present model implements a simplified version of this concept: robot transparency determines the probability that a human recognizes a task-robot mismatch, effectively operationalizing Level 2 transparency (why the robot may not be appropriate for this task).

2.3.2 Adaptive Transparency

Kunze et al. (2024) showed that adaptive transparency based on cognitive load improved performance by 32%, demonstrating that static transparency levels are suboptimal. While the present model uses fixed transparency parameters within each simulation run, the adaptive Kalman filter's stress-modulated measurement noise captures a related dynamic: under high stress (a proxy for cognitive load), humans become less responsive to transparency information, effectively reducing its functional impact even when the information is available.

2.3.3 Transparency and Trust Calibration

Shah et al. (2011) demonstrated that human-inspired plan execution with transparent communication of robot intentions improved human-robot team performance. Their work established that transparency operates not only through trust (increasing it) but also through coordination (reducing misallocation of effort). The present model's finding that transparency reduces misallocation without affecting aggregate performance is consistent with this dual pathway — transparency improves allocation quality (coordination benefit) while the performance pathway is masked by system-level compensation.

2.4 Agent-Based Modeling in Human-Robot Teams

2.4.1 ABM Methodology

Agent-based models capture emergent properties arising from individual interactions, making them well-suited for studying human-robot teams where macro-level phenomena (over-reliance, trust cascades, performance plateaus) arise from micro-level decisions (individual delegation choices). Gilbert and Troitzsch (2005) provide the methodological foundation for social simulation, establishing principles of validation and verification that the present study follows. Grimm et al. (2006) developed the ODD (Overview, Design concepts, Details) protocol for describing agent-based models, which the present study follows.

Wilensky and Rand (2015) articulate the specific advantages of ABMs for studying complex adaptive systems: the ability to model heterogeneous agents, to observe emergent phenomena that are not predictable from individual rules, and to conduct controlled experiments that are impossible in field settings. The present study leverages all three advantages — heterogeneous human agents (varying expertise, stress, trust), emergent over-reliance from simple delegation rules, and systematic parameter sweeps across reliability and transparency levels.

2.4.2 ABM in HRI

Lewis et al. (2018) demonstrated that agent-based models can predict team-level patterns in human-robot interaction with reasonable accuracy, establishing ABM as a viable methodology for HRI research. However, existing ABMs of human-robot teams generally lack two features present in the current model. First, trust-dependent task allocation — the mechanism by which trust affects team behavior beyond simple performance prediction — is absent from prior models, in which agents typically select tasks based on proximity or availability, regardless of trust state. Second, task-type sensitivity — the distinction between tasks appropriate for robot execution and those requiring human judgment — has not been modeled in the ABM context,

despite its centrality to the over-reliance problem (Parasuraman & Riley, 1997). The present model addresses both gaps, enabling the study of over-reliance as an emergent property of trust-mediated delegation decisions.

2.4.3 Validation of ABM in HRI

Sargent (2013) establishes the distinction between verification (is the model built correctly?) and validation (does the model represent reality?), further distinguishing calibration (tuning to match known data) from predictive validation (testing against held-out data). The present study achieves calibration against Hancock et al. (2021) trust trajectory benchmarks and demonstrates internal mechanism validity through polynomial model comparison and factorial experimentation. Predictive validation against independent experimental data remains a priority for future work.

2.5 Task Allocation in Human-Robot Teams

The delegation mechanism central to the present model connects to the broader task allocation literature. Gombolay et al. (2015) demonstrated that human-robot teams perform best when task allocation authority is shared between human and robot agents, rather than assigned exclusively to either. Their finding that participants preferred and performed better under robot-led scheduling challenges the assumption that human control is always optimal — a finding consistent with the present model's demonstration that human delegation decisions can be systematically suboptimal at high levels of trust.

Shah et al. (2011) developed Chaski, a human-inspired plan-execution system that improved human-robot team performance by transparently communicating scheduling decisions. Their work established that the quality of task allocation — not just the volume of tasks completed — affects team effectiveness, anticipating the present study's finding that misallocation rate captures information invisible to aggregate performance metrics.

2.6 Summary and Research Gaps

The literature establishes that trust is the primary mediator of human reliance on robotic teammates, with robot performance as the dominant antecedent. Three gaps motivate the present work:

First, while computational trust models exist (Guo & Yang, 2021; Xu & Dudek, 2015), none have been embedded in agent-based team simulations with trust-dependent task allocation. Trust

models predict trust levels; they do not predict team performance consequences of trust-mediated delegation decisions.

Second, the over-reliance problem (Parasuraman & Riley, 1997; Parasuraman & Manzey, 2010) is well-documented empirically but has not been reproduced computationally in a team simulation with task-type sensitivity. Existing ABMs assign tasks without considering whether they are appropriate for the assigned agent type.

Third, the relationship between transparency and allocation quality — as distinct from aggregate performance — has not been systematically investigated. The present model's finding that these metrics reverse predictor dominance suggests that prior studies evaluating transparency using only aggregate metrics may have underestimated its functional importance.

3. Model Design

3.1 Conceptual Framework

The model includes three interacting systems. Human agents' trust levels change based on observed robot performance, using an adaptive Kalman filter. Trust influences the willingness to delegate tasks through a task-type-sensitive delegation function. Stress builds up with workload and reduces both performance and the ability to update trust.

Robot agents operate with fixed levels of reliability and transparency. They communicate status updates probabilistically, with each communication event increasing nearby humans' trust in the robot in proportion to the robot's transparency level. Tasks are generated in two types: routine tasks (60%) for which robots have a performance advantage (+20% modifier), and judgment tasks (40%) for which humans have a performance advantage (+30% modifier). This task-type distinction is central to the over-reliance mechanism.

Figure 1 illustrates the conceptual framework of the trust-delegation-performance architecture. Robot reliability and transparency enter the adaptive Kalman trust filter, where stress and transparency modulate the measurement noise R(t). Trust drives delegation decisions through task-type-sensitive probability functions (linear for routine, quadratic for judgment tasks). A transparency-gated awareness check can block inappropriate delegation of judgment tasks. Task outcomes asymmetrically update trust (+1 success, −5 failure).

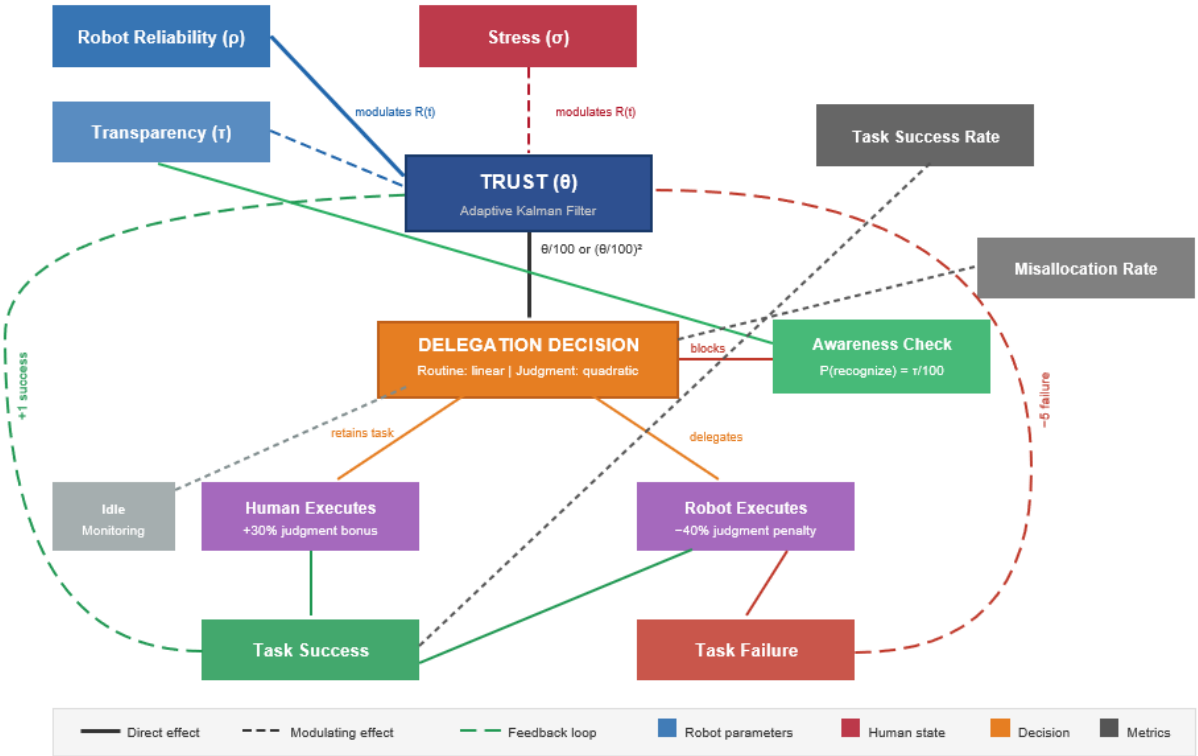

Figure 1. Conceptual Framework: Trust-Delegation-Performance Architecture.

3.2 Core Dynamics

Trust Evolution. Human trust updates follow an adaptive Kalman filter formulation with stress- and transparency-modulated measurement noise (see Appendix A for full derivation):

$$K_i(t) = P_i(t) / [P_i(t) + R_i(t)]$$

$$\theta_i(t+1) = \theta_i(t) + K_i(t) \cdot [\rho_observed - \theta_i(t)]$$

$$P_i(t+1) = (1 - K_i(t)) \cdot P_i(t) + Q$$

where the measurement noise $R_i(t)$ is adaptively modulated as:

$$R_i(t) = R_base \times (1 + \sigma^2) \times (1 + (1 - \tau)^2)$$

with σ = stress/100, τ = transparency/100, R_base = 10, and Q = 0.5. Under low stress and high transparency, R(t) is small, and the Kalman gain K(t) is high, allowing rapid trust updating. Under high stress or low transparency, measurement noise increases, and trust becomes resistant to change, leading to path dependence. Trust is additionally updated upon task completion: +1

for successes (with transparency bonus up to +1.5) and −5 for failures (buffered by transparency down to −2.5), with an extra +1/−2 for delegated tasks specifically. Trust is bounded on [0, 100].

Trust-Dependent Delegation. For routine tasks (type = 0), delegation probability is linear in trust:

$$P(delegate \mid routine) = (\theta/100) \times \delta$$

For judgment tasks (type = 1), delegation probability is quadratic:

$$P(delegate \mid judgment) = (\theta/100)^2 \times \delta$$

where δ is the delegation-sensitivity parameter. The quadratic function ensures judgment task delegation remains suppressed at moderate trust but accelerates at high trust. A transparency-based awareness check gates delegation: P(recognize mismatch) = τ/100. When recognized, the delegation is blocked. High trust + low transparency = over-reliance; high trust + high transparency = appropriate reliance.

Stress Dynamics. Stress follows bounded growth with proportional decay:

$$\sigma(t+1) = \sigma(t) + \gamma \cdot w(t) \cdot [1 - \sigma(t)/100] - \lambda \cdot \sigma(t)/100$$

where γ = 0.1 (active work) or γ_m = 0.05 (idle monitoring), and λ = 0.5 (stress decay rate). The monitoring rate reflects reduced but non-zero cognitive load during supervision of delegated tasks. Steady-state stress under constant workload is σ* = 100γw / (γw + δ).

Task Success. For humans: P(success) = min(1, ε/d) × modifier, where modifier is ×1.3 for judgment tasks and ×0.85 for routine. For robots: P(success) = (ρ/100) × min(1, ψ/d) × modifier, where modifier is ×1.2 for routine and ×0.6 for judgment tasks. Work rate is ε/50 for humans (×0.8 if stress > 70) and ψ/50 for robots (×0.7 if battery < 30).

3.3 Over-Reliance Mechanism

The model's main theoretical mechanism is the emergence of over-reliance resulting from the interaction of trust levels, delegation behavior, and task-type mismatch. When trust is low, humans handle most tasks. As trust increases, the linear delegation function initially drives the delegation of routine tasks. Since judgment task delegation depends quadratically on trust, it remains limited until trust surpasses approximately 70, after which it accelerates rapidly.

Transparent robots communicate their limitations through the awareness check, but at transparency = 50%, half of judgment delegations proceed without review. Delegated judgment tasks face a 40% robot performance penalty, leading to failures that humans would likely avoid. The performance plateau at high reliability naturally emerges without explicit programming — it is an emergent property of the interaction between these mechanisms.

4. Implementation

4.1 NetLogo Environment

The model was implemented in NetLogo 6.4.0. The simulation operates on a 33 × 33 continuous space where agents move and interact. Each tick represents one minute of simulated operation. The simulation loop proceeds: (1) update agent states including stress and battery, (2) humans find tasks with trust-dependent delegation decisions, (3) robots find available tasks using proximity-based matching, (4) execute collaborative matching, (5) process robot-to-human communication, (6) update trust via adaptive Kalman filter, (7) complete tasks with success probability evaluation, and (8) calculate metrics. Tasks replenish every 100 ticks when the count falls below the initial level.

4.2 Parameter Calibration

Model parameters were calibrated against published findings. The adaptive Kalman filter parameters (R_base = 10, Q = 0.5) yield trust trajectories consistent with those of Hancock et al. (2011, 2021). Stress parameters ($\gamma = 0.1$, $\delta = 0.5$) were calibrated to Dehais et al. (2011) and Kruijff et al. (2014). The trust asymmetry ratio (+1/−5) is consistent with negativity bias (Slovic, 1993; Baumeister et al., 2001). The delegation sensitivity parameter ($\delta = 0.5$) was set at the midpoint to allow full-range investigation in parameter sweeps.

4.3 Code Availability

The complete implementation is available at https://github.com/rkallur2/trust_calib. The repository includes the NetLogo model file (Trustv7.nlogo), BehaviorSpace experiment configurations, R analysis scripts, and all simulation output data.

5. Calibration Assessment

Trust dynamics were validated against Hancock et al.'s (2021) meta-analytic framework using a time-series baseline experiment (20 replications, 2000 ticks, reliability = 90%). Mean trust

trajectories were fitted with a saturating exponential: $\text{Trust}(t) = A - B \cdot \exp(-k \cdot t)$. The model achieved $R^2 = 0.850$ with an asymptote of 90.5, indicating trust converges toward a value consistent with the robot's actual reliability. The stress threshold is implemented at 70 on a 0–100 scale, consistent with Kruijff et al. (2014). The model does not reproduce Chen et al.'s (2018) finding that transparency improves aggregate task success by ~27%; however, as documented in Section 6, transparency significantly affects allocation quality through a different pathway.

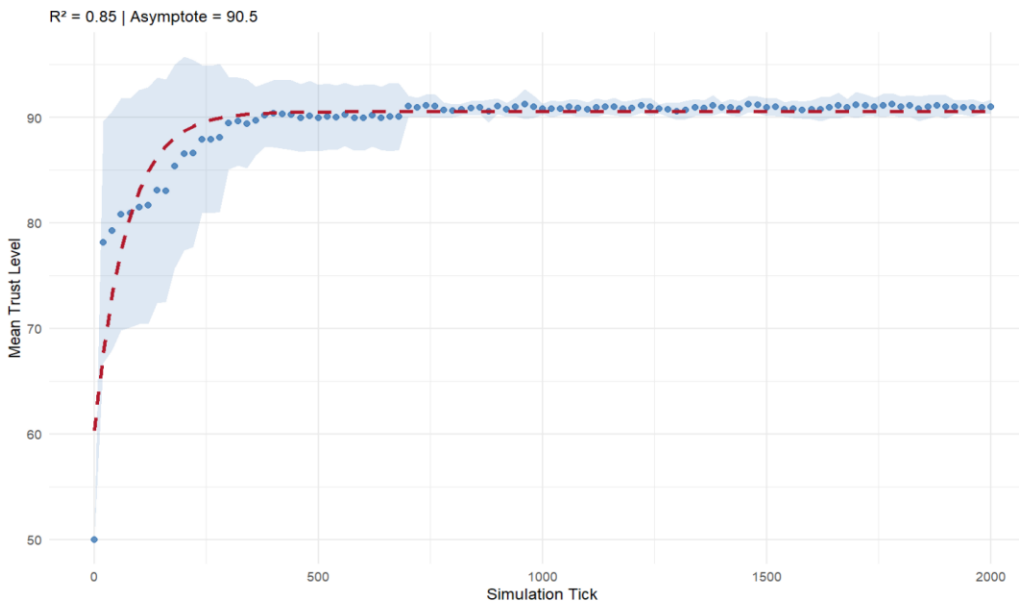

Figure 2. Trust Trajectory with Exponential Fit (Hancock Calibration).

As shown in Figure 2, mean trust follows a saturating exponential convergence toward an asymptote of 90.5, consistent with the robot's actual reliability level. The exponential fit (dashed line, $R^2 = 0.850$) confirms calibration against Hancock et al.'s (2021) meta-analytic trust development pattern. The shaded band represents ±1 SD across replications.

6. Results

6.1 Experimental Design

Model behavior was examined through three experiments:

Reliability sweep. Robot reliability varied from 30% to 100% in 5-point increments (15 levels × 50 replications = 750 runs, 2000 ticks each), all other parameters at baseline (3 humans, 3 robots,

initial-trust = 50, autonomy = 70, transparency = 50, comm-frequency = 50, delegation-sensitivity = 0.5, collaboration-rate = 50).

Reliability × Transparency. A 5 × 5 factorial crossing reliability (40, 55, 70, 85, 100) with transparency (10, 30, 50, 70, 90) at 50 replications per cell (1,250 runs, 2000 ticks each).

Trust trajectory baseline. 20 replications, 2000 ticks, reliability = 90%, metrics every 20 ticks for Hancock calibration.

6.2 Non-Linear Reliability Dynamics

6.2.1 Reliability-Performance Relationship

Robot reliability explained 85% of the variance in task success rate (one-way ANOVA: $\eta^2 = 0.851$, $F(14, 735) = 299.4$, $p < .001$). Performance increased from 55.7% (*SD* = 4.5) at 30% reliability to 85.6% (*SD* = 2.9) at 95%, before declining to 85.0% (*SD* = 3.2) at 100%. The peak below maximum reliability confirms the relationship is non-monotonic.

Polynomial model comparison (Table 1) identified a cubic polynomial as the most parsimonious (ΔAIC = 79.0 relative to the linear model; adjusted $R^2 = 0.848$). Orders 4–6 provided no improvement (ΔAIC ≤ 3.1). The relationship is predominantly linear (adjusted $R^2 = 0.831$), with a significant non-linear component adding 1.7 percentage points. The plateau above 90% reliability is robust to the choice of polynomial order from order 3 onward.

Table 1. Polynomial Order Comparison for Reliability-Performance Relationship.

Order	AIC	Adj. R^2	ΔAIC	Interpretation
1 (linear)	4406.7	0.831	79.0	Misses non-linearity
2 (quadratic)	4349.8	0.843	22.1	Captures plateau
3 (cubic)	4327.7	0.848	0.0	Best parsimonious
4	4329.4	0.848	1.7	No improvement
5	4329.4	0.848	1.7	No improvement
6	4330.8	0.848	3.1	No improvement

Note. N = 750 (15 levels × 50 replications). ΔAIC relative to best model (order 3).

Figure 3 shows the effect of reliability on Team Dynamics. Task success rate peaks at 95% reliability before declining, consistent with the cubic polynomial fit (ΔAIC = 79 relative to the linear fit). As reliability increases, trust rises monotonically, driving higher delegation rates. Misallocation rate — the proportion of delegated tasks that are judgment tasks — increases from

4.7% to 28.8%, revealing the over-reliance mechanism: high trust leads to inappropriate delegation of tasks for which robots have a 40% performance penalty. The error bars represent 95% confidence intervals.

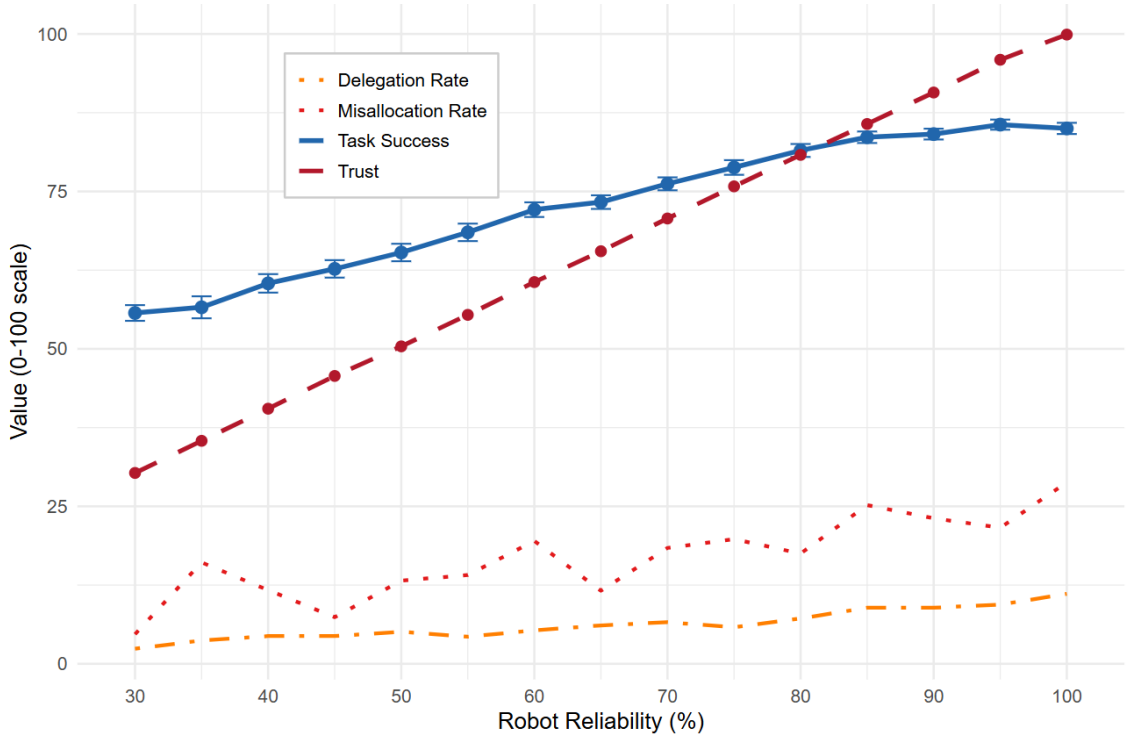

Figure 3. Effect of Robot Reliability on Team Dynamics.

6.2.2 Mechanistic Decomposition

The non-linear pattern is driven by trust-mediated delegation, producing systematic misallocation of judgment tasks. As reliability increases, trust rises (30.3 → 99.9), delegation rates increase (2.4% → 11.1%), and the proportion of delegated tasks that are judgment tasks rises from 4.7% to 28.8%. This occurs because the quadratic delegation function for judgment tasks accelerates at high levels of trust. The transparency-based awareness check ($P = \tau/100 = 0.5$ at baseline) blocks only half of inappropriate delegations. Concurrently, humans shift to idle monitoring (idle rate: 2.7% → 8.7%), partially offsetting robot performance gains. The causal chain: ↑Reliability → ↑Trust → ↑Delegation → ↑Judgment misallocation → ↑Robot failures on misallocated tasks → Performance plateau.

6.2.3 Transparency and Allocation Quality

The reliability × transparency experiment (1,250 runs) tested the moderating role of transparency. Two-way ANOVA on task success showed reliability dominance ($\eta^2 = 0.842$, $F = 1665.9$, $p < .001$), no transparency main effect ($\eta^2 < 0.001$, $p = .44$), and a non-significant interaction ($\eta^2 = 0.003$, $p = .06$).

However, the same design on the misallocation rate reversed predictor dominance entirely. Transparency became the dominant predictor ($\eta^2 = 0.090$, $F = 32.3$, $p < .001$), with reliability secondary ($\eta^2 = 0.043$, $F = 15.3$, $p < .001$). At reliability = 100%, misallocation dropped from 36.2% (T=10%) to 7.7% (T=90%), a nearly five-fold reduction as Table 2 shows.

Table 2. Contrasting Effect Sizes Across Performance and Process Metrics.

Source	Task Success η^2	Misallocation η^2	Implication
Reliability	0.842***	0.043***	Drives outcomes
Transparency	< 0.001	0.090***	Drives process quality
Interaction	0.003	0.014	Not significant
Residual	0.155	0.853	

Note. *** $p < .001$. N = 1,250 (5 × 5 × 50). η^2 from two-way ANOVA.

Figure 4 shows the effect of reliability and transparency level on the task success rate. The overlapping lines demonstrate that transparency has no detectable effect on aggregate performance ($\eta^2 < 0.001$, $p = .44$), with task success varying by fewer than 2 percentage points across transparency levels within each reliability condition. The error bars represent 95% confidence intervals.

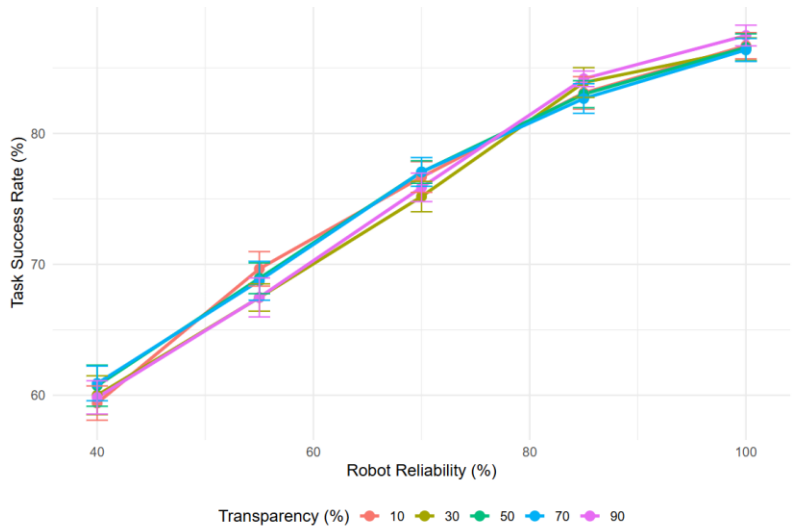

Figure 4. Task Success Rate by Reliability and Transparency Level.

Figure 5 shows the effect of robots' reliability and transparency on the task misallocation rate, using the data described in Table 3. Transparency substantially separates the curves: at reliability = 100%, misallocation drops from 36.2% (transparency = 10%) to 7.7% (transparency = 90%). This contrasts sharply with Figure 4, where the same transparency levels produce overlapping task performance curves. The pattern confirms that transparency operates on allocation quality ($\eta^2 = 0.090$) rather than aggregate performance ($\eta^2 < 0.001$), with system-level compensation masking the mechanism's effect on outcome metrics. The error bars represent 95% confidence intervals.

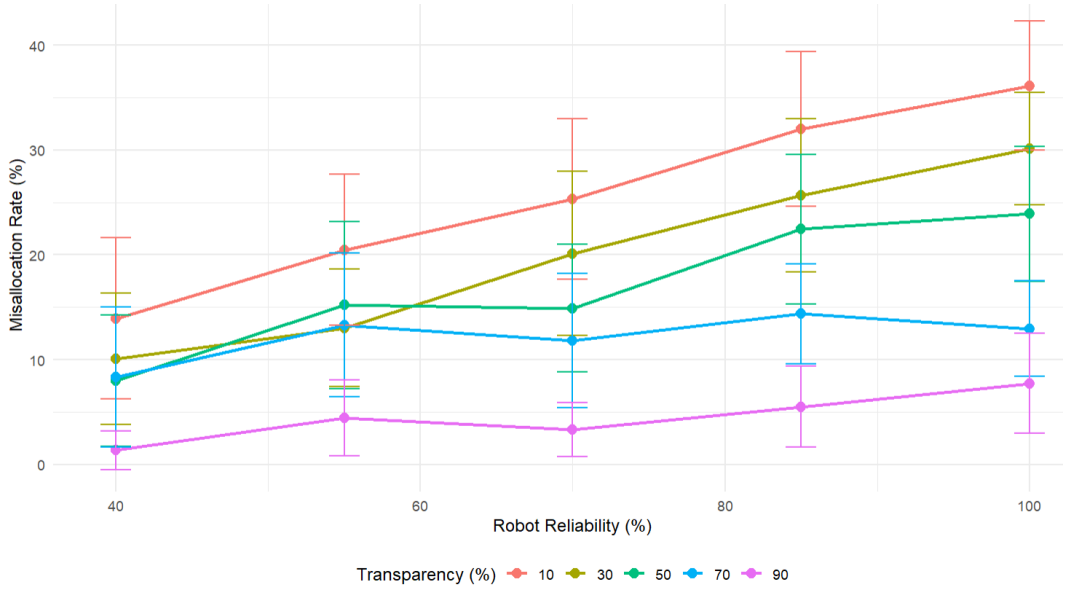

Figure 5. Misallocation Rate by Reliability and Transparency Level.

Table 3. Misallocation Rate (%) by Reliability and Transparency Level.

Reliability	T = 10%	T = 30%	T = 50%	T = 70%	T = 90%
40%	13.9	10.1	8.0	8.3	1.3
55%	20.5	13.0	15.2	13.3	4.4
70%	25.3	20.1	14.9	11.8	3.3
85%	32.0	25.7	22.4	14.4	5.5
100%	36.2	30.1	23.9	13.0	7.7

Note. T = robot transparency. Values are means across 50 replications.

This metric-dependent reversal constitutes a key finding. Transparency significantly reduces misallocation of judgment tasks, but it does not translate into differences in aggregate performance. The compensation pathway: when transparency blocks a judgment delegation, the human retains the task while the robot picks up an alternative. Net completion remains constant. Standard performance metrics, such as task success rate and throughput, may be insufficient to detect differences in allocation quality that matter for safety, accountability, and the maintenance of human skills.

7. Discussion

7.1 Theoretical Contributions

The findings address the three research gaps identified in Section 2.6. The first gap — the absence of trust-dependent task allocation in agent-based team simulations — is addressed by implementing quadratic delegation functions that produce emergent over-reliance. The second gap — the lack of computational reproduction of over-reliance with task-type sensitivity — is addressed by the non-linear reliability-performance finding driven by misallocation of judgment tasks. The third gap — the unexamined relationship between transparency and allocation quality as distinct from aggregate performance — is addressed by the metric-dependent predictor reversal finding.

Non-linear trust-performance dynamics. The model demonstrates that trust-dependent delegation produces emergent non-linear dynamics. The performance plateau near 95% reliability, driven by misallocation of judgment tasks, is consistent with Parasuraman and Riley's (1997) automation misuse framework but provides a mechanistic account through task-type-specific delegation functions. The cubic polynomial fit (ΔAIC = 79) confirms that linear models

miss the essential structure. An earlier model version, lacking trust-dependent delegation, produced purely linear results, demonstrating that the delegation mechanism is necessary and sufficient for nonlinearity.

Metric-dependent predictor reversal. The finding that transparency dominates misallocation ($\eta^2 = 0.090$) while having no effect on performance ($\eta^2 < 0.001$) reveals a measurement challenge. If evaluated solely by task success rate, transparency appears irrelevant. Misallocation rate tells a different story entirely. This has implications for research methodology: evaluations should include process metrics (delegation appropriateness, task-type matching) alongside outcome metrics. The finding also extends the literature on AI over-reliance by demonstrating that system-level compensation can mask the effects of local mechanisms.

Over-reliance as an emergent property. The over-reliance dynamic arises from interactions among mechanisms rather than being programmed directly. This illustrates the value of agent-based modeling for studying phenomena arising from micro-level decisions — consistent with the bottom-up approach advocated in the ABM literature (Gilbert & Troitzsch, 2005).

7.2 Practical Implications

Reliability remains the dominant design lever ($\eta^2 = 0.851$), but the plateau above 95% suggests diminishing returns from marginal reliability investments at high levels. Transparency's role is more nuanced than aggregate metrics suggest: it substantially reduces misallocation of judgment-critical tasks, a distinction with direct implications for safety-critical domains. Team designers should complement performance metrics with measures of allocation quality.

7.3 Limitations

The model has not been validated against independent experimental data. The transparency mechanism affects allocation quality but not performance, which may reflect emergent compensation or insufficient mechanism strength. Human agents do not learn or adapt strategies. Robot reliability is fixed. Stress dynamics do not respond to robot failures. The model has not undergone formal face validity evaluation by independent experts. The delegation sensitivity parameter was held at 0.5; systematic variation could reveal whether higher sensitivity amplifies the plateau effect.

7.4 Future Work

The most immediate priority is out-of-sample validation. The model generates testable predictions: a performance plateau near 95% reliability, misallocation increasing with trust, and transparency reducing misallocation without affecting throughput. Second, the reliability × delegation interaction should be examined to test whether higher delegation sensitivity amplifies over-reliance. Third, variable reliability (rather than fixed) would enable the study of trust repair after transient failures. Additional extensions include agent learning, larger teams, and task dependencies.

8. Conclusion

This paper presents an agent-based framework with trust-dependent delegation for studying trust dynamics and performance in human-robot teams. Three principal findings emerge. First, the reliability-performance relationship is non-linear (cubic, $\Delta AIC = 79$; $\eta^2 = 0.851$), with performance peaking at 95% due to trust-mediated misallocation of judgment tasks. Second, transparency significantly reduces misallocation ($\eta^2 = 0.090$) without affecting aggregate task success ($\eta^2 < 0.001$), indicating that standard performance metrics mask differences in allocation quality. Third, trust trajectories match meta-analytic benchmarks ($R^2 = 0.850$). Together, these findings address the three gaps identified in the literature: the absence of trust-dependent delegation in team simulations, the lack of computational over-reliance models, and the unexamined transparency–allocation quality relationship. These results demonstrate that trust-dependent delegation is necessary for over-reliance to emerge. The open-source implementation provides testable predictions for experimental validation.

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During the preparation of this work, the author used a generative AI tool to improve grammar and writing style. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the publication's content.

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Availability of Data

The agent-based model, simulation data, and statistical analysis scripts are available athttps://github.com/rkallur2/trust_calib under a Creative Commons license.

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Appendix A: Mathematical Framework

A.1 Notation

$\theta_i(t) \in [0,100]$: Trust of human i at time t. $\varepsilon_i \in [50,99]$: Expertise (fixed). $\sigma_i(t) \in [0,100]$: Stress. $w_i(t) \in \mathbb{R}^+$: Workload. $\rho_j \in [0,100]$: Robot reliability. $\tau_j \in [0,100]$: Transparency. $\psi_j \in [60,99]$: Capability. $d_k \in [10,99]$: Task difficulty. $\delta \in [0,1]$: Delegation sensitivity.

A.2 Adaptive Kalman Trust Filter

The trust update implements a Kalman filter with adaptive measurement noise. At each tick, for human i observing nearby robots:

Step 1: Compute observed performance. ρ_obs = mean reliability of robots within radius 10.

Step 2: Compute adaptive measurement noise.

$$R_i(t) = R_base \times (1 + (\sigma_i/100)^2) \times (1 + (1 - \bar{\tau}/100)^2)$$

where R_base = 10, σ_i is current stress, and $\bar{\tau}$ is mean transparency of nearby robots. The squared terms ensure non-linear sensitivity: R grows slowly for moderate stress/low-transparency but accelerates at extremes. At $\sigma = 0$, $\tau = 100$: R = 10. At $\sigma = 100$, $\tau = 0$: R = 10 × 2 × 2 = 40.

Step 3: Compute Kalman gain.

$$K_i(t) = P_i(t) / [P_i(t) + R_i(t)]$$

where $P_i(t)$ is the trust covariance (uncertainty), initialized at 25 and bounded on [0.1, 50].

Step 4: Update trust state.

$$\theta_i(t+1) = \theta_i(t) + K_i(t) \cdot [\rho_obs - \theta_i(t)]$$

Step 5: Update covariance.

$$P_i(t+1) = (1 - K_i(t)) \cdot P_i(t) + Q$$

where Q = 0.5 is the process noise, representing inherent uncertainty in the trust evolution process. The covariance P tracks trust uncertainty: high P means the agent is uncertain and updates rapidly; low P means the agent is confident and updates slowly.

A.3 Task Completion Trust Updates

Upon task completion by a robot observed by nearby humans:

Success: $\Delta\theta = +1 + \tau/200$ (transparency bonus up to +1.5)

Failure: $\Delta\theta = -5 \times (1 - \tau/200)$ (transparency buffer reduces to −2.5)

Delegated task bonus: +1 additional on success, −2 additional on failure, applied to the delegating human specifically.

A.4 Trust-Dependent Delegation

When human i encounters available task k:

$$P(delegate \mid type=0) = (\theta_i/100) \times \delta$$
$$P(delegate \mid type=1) = (\theta_i/100)^2 \times \delta$$

If delegation is attempted for a judgment task (type=1), a transparency check occurs:

$$P(recognize\ mismatch) = \tau_robot / 100$$

If recognized, delegation is blocked and the human retains the task. Additionally, a capability check blocks delegation if task difficulty exceeds robot capability by more than 10 points.

A.5 Stress Dynamics

Stress evolves via bounded growth with proportional decay:

$$\sigma(t+1) = \sigma(t) + [\gamma_eff \cdot w(t) \cdot (1 - \sigma(t)/100) - \lambda \cdot \sigma(t)/100] \times 100$$

where $\gamma_eff = 0.1$ for active work, $\gamma_eff = 0.05$ for idle monitoring, and $\lambda = 0.5$ *for stress decay rate*. Setting $d\sigma/dt = 0$:

$$\sigma^* = 100 \cdot \gamma_eff \cdot w / (\gamma_eff \cdot w + \lambda)$$

For typical workload w = 5 with active rate: $\sigma^* = 100 \times 0.5 / (0.5 + 0.5) = 50$. The saturation term $(1 - \sigma/100)$ ensures stress asymptotically approaches the ceiling.

A.6 Task Success Probability

Human success:

$$P_h = min(1, \varepsilon/d) \times m_h(type)$$

where m_h(0) = 0.85 (routine penalty), m_h(1) = 1.30 (judgment bonus).

Robot success:

$$P_r = (\rho/100) \times min(1, \psi/d) \times m_r(type)$$

where m_r(0) = 1.20 (routine bonus), m_r(1) = 0.60 (judgment penalty).

Work rate: $\omega_h = \varepsilon/50$ ($\times 0.8$ if $\sigma > 70$). $\omega_r = \psi/50$ ($\times 0.7$ if battery < 30).

A.7 Team Efficiency

$$E = |\{agents\ with\ assigned\text{-}task \neq nobody\ or\ idle\text{-}monitoring\}| \ / \ |all\ agents|$$